

College of Computing

Department of Software Engineering

Course Title: WEB DEVELOPMENT ll

PROJECT TITLE :The Millennium School E-Learning System

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The Millennium School E-Learning System

1.Overview

The Millennium School E-Learning System is a comprehensive web-based Learning Management System (LMS) and academic administration platform designed to streamline teaching, learning, and school management processes. The system provides a centralized digital environment where administrators, teachers, students, and parents can collaborate through role-based dashboards tailored to their responsibilities.

Built with a **custom PHP REST API backend** and a **React/Vite frontend**, the platform supports digital course delivery, assignment management, attendance tracking, performance analytics, communication tools, and parental engagement. It enables educational institutions to efficiently manage academic activities while improving transparency and student performance monitoring.

2.Key Features

Academic Management

- Manage classes, departments, courses, and academic structures.
- Assign teachers to courses and classes.
- Maintain student and teacher records.
- Organize learning content by grade, subject, chapter, and topic.

Digital Learning Platform

- Upload and distribute learning materials.
- Support multimedia content and file storage through Cloudinary integration.
- Provide structured course navigation and content access for students.

Assignment & Assessment Management

- Create and publish assignments.
- Accept online student submissions.
- Review, grade, and provide feedback.
- Track submission status, deadlines, and grading history.

Attendance Management

- Bulk attendance marking for classes.
- Student attendance history and monthly summaries.
- Course-level and institution-wide attendance analytics.

- Export attendance reports in CSV format.
- Identify attendance-related risk indicators.

Performance Analytics & Student Monitoring

- Academic performance dashboards.
- Course progress tracking and trend analysis.
- Student risk assessment and early-warning indicators.
- Learning gap analysis and performance reporting.

Communication & Collaboration

- Course discussion groups and forums.
- Announcements and school-wide notifications.
- Real-time communication support through Server-Sent Events (SSE).
- Optional broadcasting integrations for live updates.

Parent Portal

- Monitor children's academic progress.
- View attendance records and assignment status.
- Access performance summaries and risk reports.
- Receive notifications and academic updates.

Help & Support Center

- Frequently Asked Questions (FAQs).
- User guides and documentation.
- Support ticket management system.
- Ticket responses and notification tracking.

3.User Roles

Administrator

Administrators oversee the entire platform, including user management, academic structure configuration, reporting, analytics, attendance monitoring, and system administration.

Teacher

Teachers manage courses, upload learning materials, create assignments, grade submissions, record attendance, participate in discussions, and monitor student performance.

Student

Students access course materials, submit assignments, participate in discussions, track attendance, view grades, and receive notifications.

Parent

Parents monitor their children's academic activities, attendance, performance reports, assignments, and school communications.

4.System Architecture

The platform follows a modern client-server architecture:

Backend

A custom PHP-based REST API handles business logic, authentication, database operations, reporting, and academic workflows.

Frontend

A React-based Single Page Application (SPA) consumes API endpoints and provides responsive, role-specific user interfaces.

Database

MySQL stores academic records, user data, assignments, attendance information, notifications, discussions, and reporting data.

5.Technology Stack

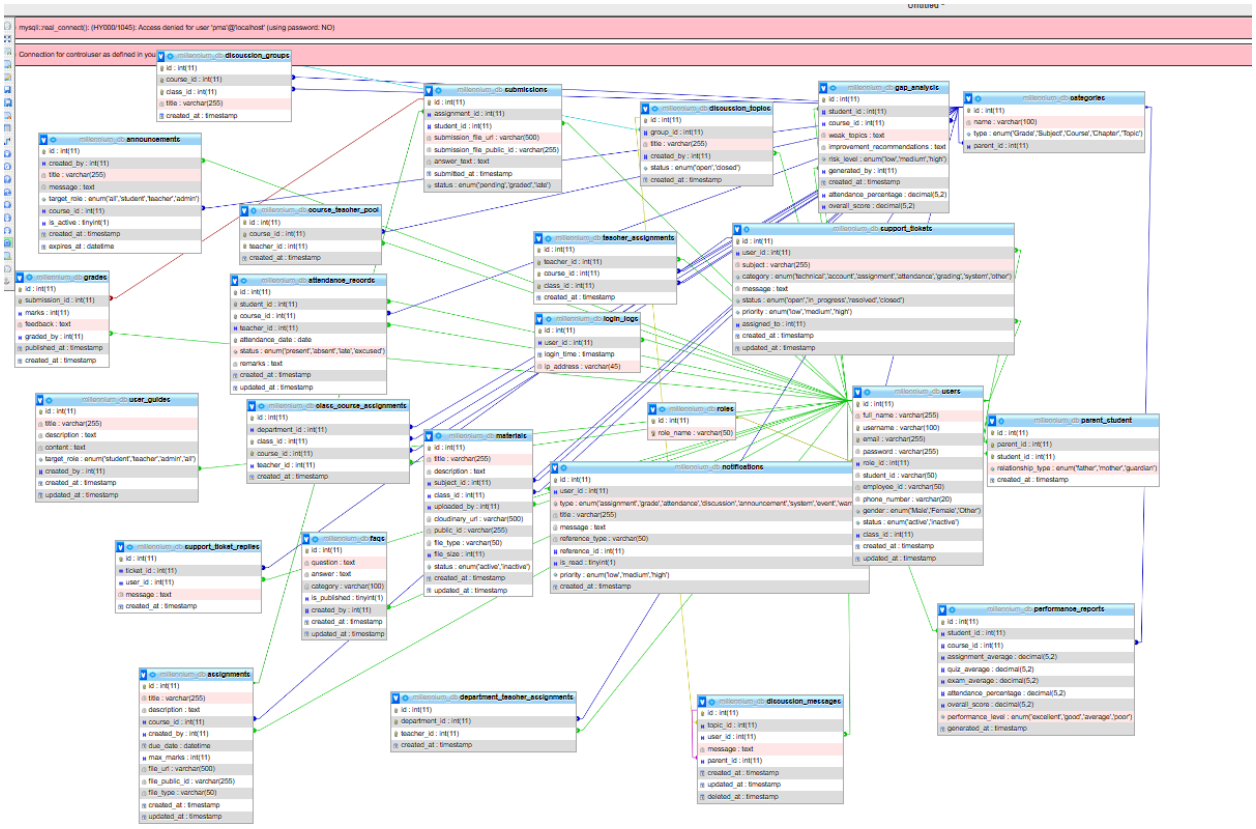
Backend

- PHP 8+
- Custom MVC-inspired Architecture
- RESTful API Design
- Composer & PSR-4 Autoloading
- MySQL
- Authentication Middleware
- JWT/Session-Based Security
- Cloudinary Integration
- Optional Real-Time Broadcasting

Frontend

- React
- Vite
- React Router
- TanStack React Query
- Axios
- Tailwind CSS
- Framer Motion
- Recharts
- Lucide React Icons
- Pusher JS

6.Database Design



7.Core Modules

- Authentication & Authorization
- User Management
- Academic Administration
- Course Management
- Learning Material Delivery
- Assignment & Submission Management
- Grading & Assessment
- Attendance Tracking
- Performance Analytics
- Risk Detection & Gap Analysis
- Discussions & Collaboration
- Announcements & Notifications
- Help Center & Support Tickets
- Parent Monitoring Portal

8.Key Benefits

- Centralized academic management platform.
- Improved communication among administrators, teachers, students, and parents.
- Enhanced transparency through performance and attendance tracking.
- Early identification of at-risk students through analytics.
- Scalable architecture suitable for schools and educational institutions.
- Easy deployment using standard PHP and MySQL hosting environments.

Conclusion

The Millennium School E-Learning System provides a complete digital ecosystem for modern educational institutions. By integrating learning management, academic administration, analytics, communication, and parental involvement into a single platform, it helps schools improve operational efficiency, enhance student engagement, and support data-driven educational decisions.